

this reason, DNA samples, odor etc. are not taken into consideration. Some of the traits, like speech data, though represented directly as signals, may be converted to image form for suitable analysis.

Face, fingerprint, palmprint, iris, ear and several others are the different identifiers used for the purpose of automatic human recognition. Each of them carries the pros and cons of their own. Seven characteristics [25], namely universality, distinctiveness, permanence, collectability, performance, acceptability and circumvention should be prioritized as per researcher's goal before the workflow design begins. For example, if permanence is assigned more priority than distinctiveness, then ear should be a preferred choice over face. Similarly, if performance is the key, iris can be an automatic alternative. However, choice of a biometric identifier largely depends upon the researcher's interest, research specific goal and availability of concerned databases.

4. Workflow pattern for a MMBs

MMBS are based on capturing human bodily features and using them for identification and authentication. MMBs function in an abstract manner. Generally, CBIR (Content Based Image Retrieval) system retrieves a query image and primitive features denoting image content, such as colour, textures, and shape, are computed for both stored and query images, and then used to identify stored images, most closely matching the query image. Semantic features such as the type of object present in the image, though difficult to extract, remains an active research area. Similarly, in a MMBs a query image, showing a biometric trait (say, face or fingerprint or iris etc.) is searched from a stored database of the same biometric. An overview of the generic MMBs is illustrated with the help of Fig.2. The entire procedure is divided into two sub-phases, namely training and testing.

4.1. Training phase in workflow

During the training phase, let us consider that the input database consists of multiple biometric modalities and is assumed to be structured and known. The term known indicates that the system and by structured we mean that every instance is properly placed within its class once enrolled. For example, every fingerprint of a person X_j should belong to class j , considering the presence of any person X_i in a database of n persons, where $i = 1, 2, \dots, j, \dots, n$.

The main algorithm of the system used to match a known and an unknown instance is called the matcher. The matcher is used to classify an unknown instance to its proper class. The efficacy of a matcher is adjudged by its score, crudely a number of correct classifications in comparison to wrong ones. The score can be viewed as the rate of recognition of the system. The different input

parameters of a matcher, for example, a global classification threshold value, can be adjusted at the end of each epoch of the algorithm in order to produce a better result. In other words, considering the input parameters and matcher as an encapsulated entity, the matcher is as a whole modified or trained to enhance the level of score.

The dataset used to train the classification algorithm is called the training dataset. Evidently, the term “trained dataset” would be wrong as the dataset is itself not trained and rather used to train the algorithm. Other than the training part the input database is divided into what is called a validation set, the dataset used to validate the matcher. The matcher takes an instance from the validation set and classifies it. Whether it is a mismatch or a correct classification can easily be determined, since the validation set is actually designed from the known database. Based on the result of classification, several input parameters are adjusted to enhance the system outcome. Here human interactions (not automated), for example, manual manipulation of classification threshold value may take place. The training phase concludes once the developer is satisfied with the system outcome. Therefore, the entire training phase helps to train the classification algorithm to a level of significant satisfaction.

4.2. Testing phase in workflow

In the testing phase, the unknown instances are fed to the modified matcher for classification. The database used to test the accuracy of the system is the test database. A robust MMBs should not only classify a test instance (closed set scenario) but also specify if the unknown instance does not fall into any of the existing known classes (open test scenario). In both the cases, the raw data captured is normalized into feature vector form and fusions are carried out at different levels. In the preprocessing phase, one may select the Region of Interest (RoI) of the image instance for further processing. Before feeding the data to the matcher, its dimensionality is reduced for faster processing and the required features (based on which the matcher is designed), generally in vector form, are extracted. Some of the feature extraction and dimensionality reduction schemes commonly used are PCA, KPCA, 2DPCA etc.

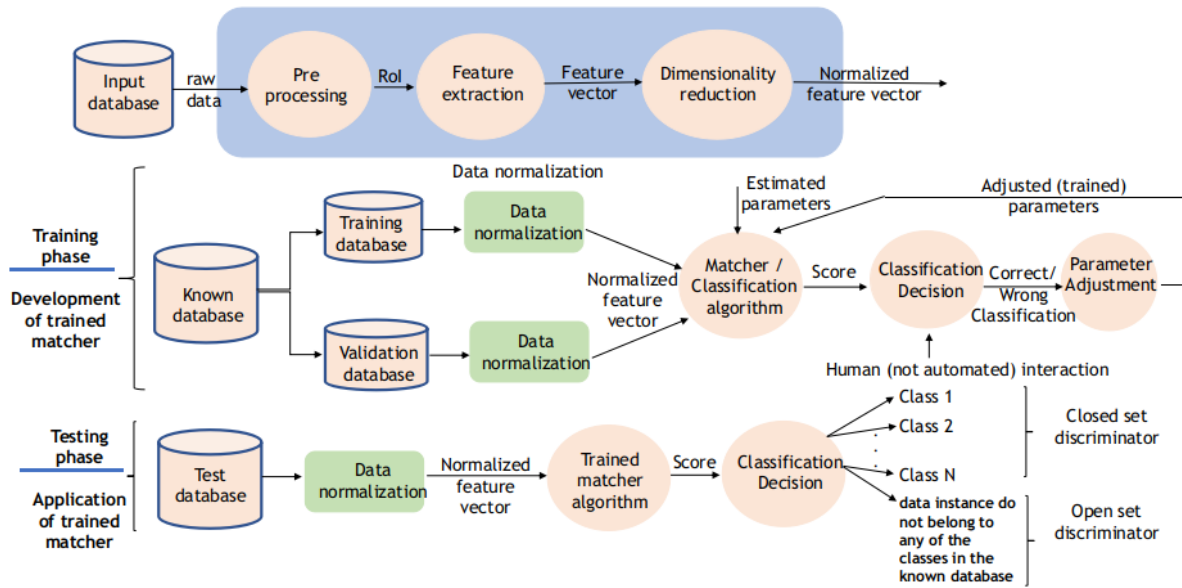


Figure 2: Abstract overview of a biometric recognition system

4.3. Workflow modes

The acquisition and processing in MMBs is carried out in three different modes.

- i. **Serial mode:** Here, a biometric trait is processed before accepting the next one and the decisive classification outcome of the former is forwarded to the latter. It may reduce the overall recognition time as a decision can be drawn without accepting the next modality in sequence.
- ii. **Parallel mode:** Here multiple modalities are processed simultaneously and the conclusion is drawn integrating the outcomes of all modalities.
- iii. **Hierarchical mode:** Here individual matchers are combined in a treelike structure in order to manage several classification approaches.

5. Levels of information fusion in workflow

In MMB scenarios, how to fuse information of two different biometrics of the same person remains a matter of research. These fusion strategies can be performed in raw level, or in feature level, or in score level or even in decision level [24]. In the raw (or sensor) level of fusion, the same characteristics of raw MMB data captured using different sensors (for example, sensing the fingerprint of the same finger of a person in more than one fingerprint scanner) are combined. At times it is found to enhance the recognition accuracy but suffers from the drawback of incompatibility of data from different modalities.